

Neel Viegas

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EDUCATION

- **McGill University**
Bachelor of Mechanical Engineering Expected December 2028
- **John Abbott College**
Science program – Dean's honor list May 2025
 - English – Honors certificate
- **College Charlemagne**
High School Pure sciences – International Baccalaureate (IB) Program June 2023
 - Certificate of exceptional performance (+90% average)

WORK EXPERIENCE

- **Pratt and Whitney Internship**
Engineering Intern for Operations Program Management May 2026 – Present
 - Performed GD&T and CAD work, adding dimensions in both the 3D and 2D views in CATIA, allowing all views and dimensions to be read directly from the model without a separate drawing
 - Dissected a PW210 engine CAD and learned engine architecture basics
 - Taught SolidWorks design fundamentals to a senior colleague for application in youth volunteer programs
 - Optimized an AI database (pre-processing, integrated new data by writing a python script that extracted data from excel and pdf files)
 - Built an interactive knowledge graph app using Python, HTML and Java
 - Built a custom Power BI Gantt chart using a Deneb/Vega JSON spec, features including: cross-filtering dropdowns, computed dependency/milestone layout, 100+ tasks
 - Built a scheduled Power Automate Planner to SharePoint flow with batched Graph calls and concurrency control
 - Created an app that automatically builds budget presentation PowerPoints from excel files using Python and Excel macros, saving a Sr Manager 1h per week
- **McGill Rocket Team**
Co-lead of the injector project May 2026 – Present
 - Leading and assigning projects to improve rocket performance
 - Testing injectors on our Injector Test Rig, building a thrust analysis code
 - Managing the project: Hosting meetings, recruiting and onboarding new members, documenting progress
- *Member of injector project* September 2025 – May 2026
 - 3D CADed vortex injector drilling jig in Siemens NX
 - Researched redesign of vortex injector to optimize oxidizer-fuel mixing in hybrid rocket propulsion system
- **Physics and Mathematics Tutor**
Private Tutor September 2023 – Present
 - Teaching complete Cegep level classes, see example class: https://www.youtube.com/watch?v=INfTMg_OgAE&t=78s
 - Answering question on Calculus 1-3, Mechanics and Electricity and Magnetism
- **CI Assante Wealth Management Internship**
Assistant to financial advisor August 2023
 - Learned basics of: DCF models, retirement planning, stock valuation

PROJECTS <https://neelviegas.github.io/>

- **Thrust Analysis Tool (McGill Rocket Team)** May 2026 – Present
 - Led the creation of a python script that outputs the coefficient of combustion efficiency
 - Used Python and mapped data across numerous sensors and platforms
- **Local AI Model (Personal Project)** June 2026 – Present
 - Building a retrieval-augmented generation pipeline using the Qwen 32B model
 - Wrote ingestion, orchestration and query scripts to help answer my thermodynamic questions for the Rocket team and Thermo 2 (Reference document: Thermodynamics for Engineers v7, ~1000 pages)
- **Automatic Curtain Opener (Personal Project)** June 2026 – July 2026
 - CADed the gears and casing in Solidworks and created an assembly
 - Iterated on the gear set until the torque and fit were right
 - Designed and assembled the circuit with an Arduino
- **Chair with Assisted Seating and Standing (MECH 292)** January 2026 – May 2026
 - CADed the assembly (alone)
 - Designed and assembled a circuit (team)
 - Presented the project to the class and answered design questions (team)
- **Vortex Injector Redesign (McGill Rocket Team)** August 2025 – May 2026
 - Presented several improvements to our injector based on new research
 - Updated the CAD for the full-scale and low-thrust version

SKILLS & INTERESTS

• CAD:	NX, Solidworks, Fusion 360, CATIA
• Coding:	Python - numpy, xldr, openpyxl, json, networkx; HTML, Java (basics)
• Microsoft:	Power Automate, Excel, Word
• Teaching:	STEM teaching, CADing, problem-solving
• Languages:	Fluent in English and French
• Interests:	Backpacking (Casablanca to Paris summer 2025); Bouldering & Powerlifting

COMMUNITY INVOLVEMENT

• Pratt and Whitney Intern Committee Representative	May 2026 – Present
◦ Helped organize intern events with committee members	
• Refugee Language Teaching	March 2026 – Present
◦ Helping refugees learn English and French on Haven	
• Club Membership	August 2025 – Present
◦ Member of the McGill climbing club	
• Student Club Management	
◦ Founder and leader of the Powerlifting Club (John Abbott College)	January 2024 - May 2024
◦ Co-leader of the Chess Club (John Abbott College)	September 2023 - December 2024